Leo Wang

Email: leokingw@andrew.cmu.edu | **Cell:** (860) 960-9539 | **Portfolio:** leo01110111.github.io/projects

EDUCATION

Carnegie Mellon University | Pittsburgh, PA
B.S. Electrical and Computer Engineering | May 2028
QPA: 3.74/4.0

PUBLICATIONS & PRESENTATIONS

- S. Kowshik, S. Venkataraman, **L. Wang**, N. Pant, M. Simchowitz, A. Kumar. *Adaptation of Generalist Robot Policies with Minimal Data*. RSS 2026 Workshop on Reinforcement Learning for Vision-Language-Action Models (RL4VLA), Jul 2026.
- **L. Wang**. *Quadrupedal Slope Climbing via Deep Reinforcement Learning*. Oral presentation, Meeting of the Minds 2026, CMU Undergraduate Research Symposium, Apr 2026.
- **L. Wang**, A. Panda, V. Chandra, G. Kaimburi. *Steering Vision Language Action Models with Mechanistic Interpretability*. Course project report, 11-785 Introduction to Deep Learning, CMU, Apr 2026.

NOTABLE PROJECTS

Dexterous Foundation Model from Egocentric Human Data | Max Simchowitz Lab, with Bosch | Sep 2026 – Present

- Set up a 10,000-hour egocentric video training pipeline for a general-purpose five-fingered hand foundation model
- Built real-world and simulated evaluation environments for dexterous hand policies
- Developed post-training retargeting pipelines mapping human hand motion to robot hands

Bimanual UR7e Teleoperation and Policy Learning Setup | Max Simchowitz Lab | May 2026 – Aug 2026

- Built bimanual UR7e teleoperation with Dynamixel leader arms for data collection, behavior cloning, and policy execution
- Wrote the Python control stack (ZeroMQ, ROS 2); configured the Universal Robots arms; designed parts in CAD
- Built a MuJoCo simulation of the arms; evaluated Pi0.5 policies on the setup

Quadrupedal Climbing Policy via Deep Reinforcement Learning | May 2025 – Present

- Achieved a policy that climbs hills of 45 degrees
- Developed the training environment in Isaac Lab
- Analyzed training runs and tuned environmental parameters to achieve better performance
- Sim2Real on the Unitree Go2 robot

Robotic Arm | Personal Project | July 2024 – Present (intermittent)

- Wrote a differential IK controller with only the Eigen library in C++
- Used ROS 2 for communication of joint states/commands.
- Designed and built a 6-degree-of-freedom robotic arm from scratch
- Simulated the custom robotic arm in Isaac Sim.

WORK EXPERIENCE

Max Simchowitz Lab | Researcher | Carnegie Mellon University | May 2026 – Present

- Head of the lab's robotic infrastructure and point of contact for all robot hardware questions; built its bimanual UR7e setup.
- Developing a five-fingered dexterous hand foundation model from egocentric data in collaboration with Bosch.

Robomechanics Lab | Researcher | Carnegie Mellon University | May 2025 – Dec 2025

- Taught a bootcamp on using the Pittsburgh Supercomputing Center's GPUs for Isaac Lab training.
- Wrote batch scripts for training models on compute clusters
- Authored how-to guides on using the compute clusters and deploying policies to the Go2 robot
- Made a program that controls the Unitree Go2 robots in MuJoCo with a keyboard

SKILLS

Software: Isaac Lab, Isaac Sim, MuJoCo, PyTorch, JAX, ROS 2, Python, C++, C, Bash, Git, Claude Code
Hardware: Onshape, Arduino Prototyping, 3D Printing, Laser Cutting

NOTABLE COURSES

- 11-785 Introduction to Deep Learning
- UC Berkeley CS 285 Deep Reinforcement Learning (online)
- MIT 6.S184 Introduction to Flow Matching and Diffusion Models (online)

LEADERSHIP EXPERIENCE

Hotchkiss Robotics Team (FIRST Tech Challenge) | Captain | 2021 – 2024

- Competed in the world championship
- 1st in Connecticut state championship
- Coded autonomous routines for the robot
- Created various mechanical systems using CAD, 3D printing, and laser cutting